

Stock_Price_Prediction_Report

1. Abstract

This project focuses on predicting stock prices using IBM SPSS Modeler. Historical stock market data was analyzed using machine learning models such as C&R Tree and Regression. The project helps in understanding stock market trends and generating accurate predictions.

2. Introduction

Stock market prediction is an important application of data mining and machine learning. Investors use prediction systems to analyze market trends and make better financial decisions. IBM SPSS Modeler provides powerful tools for predictive analytics and data visualization.

3. Objectives

- To analyze stock market data
- To predict stock prices
- To compare machine learning models
- To identify the best prediction model
- To understand predictive analytics using SPSS

4. Dataset Description

The dataset contains historical stock market information including:

- Open Price
- High Price
- Low Price
- Close Price
- Volume

The dataset was imported into IBM SPSS Modeler using the Flat File node.

5. Methodology

- Import Dataset using Flat File Node
- Define field types using Type Node
- Apply preprocessing and derive new field Profit = Close - Open
- Use Table and Histogram nodes for visualization
- Apply C&R Tree and Regression models
- Analyze prediction results

6. Models Used

- C&R Tree
- Regression

Among all models, the C&R Tree model provided the best prediction accuracy.

7. Results

The project successfully generated prediction models using IBM SPSS Modeler. The C&R Tree model achieved high prediction accuracy with a correlation value of 0.999 and very low relative error.

8. Conclusion

The Stock Price Prediction project successfully analyzed stock market data and generated accurate prediction results using IBM SPSS Modeler. Machine learning techniques helped in understanding market trends and improving prediction accuracy.